MATH105: INTRODUCTION TO STATISTICS

# PURSUING A CAREER IN DATA SCIENCE

**A STATISTICAL GUIDE FOR MATH GRADUATES**

## Introduction

This data analysis is designed to serve as a career orientation guide for the U.S. Math graduates who are considering a career in Data Science. It provides a detailed exploration of several critical aspects of the job market—including salary levels, employment duration, the relationship between company size and experience level, and work arrangements (such as on-site vs. remote work). Drawing on a comprehensive dataset and supported by relevant research findings, the analysis does not claim to offer a one-size-fits-all formula. Rather, it aims to present a credible and data-driven portfolio of recommendations to help graduates make informed decisions based on their individual goals and circumstances.

## Dataset & Sample Selection

### Dataset

This study draws on the **AI–ML–Data-Science Salary (2020 – 2025)** dataset, which records compensation for roles in artificial intelligence, machine learning, data science, and other tech-related roles. All salary figures have been converted to U.S. dollars using the average annual foreign-exchange rate for the year in which the salary was earned, ensuring comparability across currencies and time.

### Key Attributes

- **work_year** — Calendar year in which the employee received the salary (2020 – 2025)
- **experience_level** — Seniority of the employee in that year
  - EN = Entry-level / Junior
  - MI = Mid-level / Intermediate
  - SE = Senior-level / Expert
  - EX = Executive-level / Director
- **employment_type** — Nature of the employment contract:
  - FT = Full-time PT = Part-time
  - CT = Contract FL = Freelance

- **job_title** — Official title held during the reported year.
- **salary** — Gross compensation in the original currency.
- **salary_currency** — ISO 4217 code of the original currency.
- **salary_in_usd** — Salary converted to USD using the period's average FX rate.
- **employee_residence** — ISO 3166 country code for the worker's primary residence that year.
- **remote_ratio** — Proportion of work performed remotely:
  - 0 = On-site ( < 20 % remote)
  - 50 = Hybrid / Partially remote
  - 100 = Fully remote ( > 80 % remote)
- **company_location** — ISO 3166 country code of the employer's headquarters or contracting office.
- **company_size** — Average workforce size during the year:
  - S = Small (< 50 employees)
  - M = Medium (50 – 250 employees)
  - L = Large (> 250 employees)

## Sample Selection

To focus the analysis exclusively on Data-Scientist roles in the U.S market, the dataset is

filtered to retain only observations whose **job_title** equals **"Data Scientist"** and **company_location** equals **"US"**. All other entries are removed, ensuring that the resulting analytic sample reflects compensation patterns for the target audience alone.

```
# Filter only job titled data scientist and company location in the US
data_scientist <- data %>%
  filter(job_title == "Data Scientist"  & company_location == "US")
data_scientist
```

Table Chart Filter Columns Search

| Index | work_year | experience_level | employment_type | job_title | salary | salary_currency | salary_in_usd | employee_residence | remote_ratio |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2025 | SE | FT | Data Scientist | 160000 | USD | 160000 | US | |
| 2 | 2025 | SE | FT | Data Scientist | 130000 | USD | 130000 | US | |
| 3 | 2025 | MI | FT | Data Scientist | 140000 | USD | 140000 | US | |
| 4 | 2025 | MI | FT | Data Scientist | 120000 | USD | 120000 | US | |
| 5 | 2025 | SE | FT | Data Scientist | 193750 | USD | 193750 | US | |
| 6 | 2025 | SE | FT | Data Scientist | 124000 | USD | 124000 | US | |
| 7 | 2025 | SE | FT | Data Scientist | 258000 | USD | 258000 | US | |
| 8 | 2025 | SE | FT | Data Scientist | 172000 | USD | 172000 | US | |
| 9 | 2025 | MI | FT | Data Scientist | 118273 | USD | 118273 | US | |
| 10 | 2025 | MI | FT | Data Scientist | 87419 | USD | 87419 | US | |
| 11 | 2025 | SE | FT | Data Scientist | 173296 | USD | 173296 | US | |
| 12 | 2025 | SE | FT | Data Scientist | 103978 | USD | 103978 | US | |
| 13 | 2025 | SE | FT | Data Scientist | 115000 | USD | 115000 | US | |
| 14 | 2025 | SE | FT | Data Scientist | 110000 | USD | 110000 | US | |
| 15 | 2025 | SE | FT | Data Scientist | 201783 | USD | 201783 | US | |

Rows: 9,090 truncated from 11,945 rows Expand

The filtered dataset then has 11,945 observations–a good sample size to start with.

# Inference for a numerical variable

To set the stage, this section aims to compare the salary between Data Scientists and other general tech roles to see if it is financially worthwhile to pursue Data Science in the broad technology industry? Is the excitement about the job prospects for Data Science grounded in reality?

## Literature Review

To conduct this analysis, we begin by focusing on the median salary rather than the average. This choice is informed by the nature of our dataset, which covers a wide range of experience levels—entry-level, mid-level, senior, and executive—potentially making the average salary misleading. With averages, unusually high or low numbers can skew the results. For instance, multi-millionaires or billionaires might drive the average higher than what the typical worker actually makes. The median is less affected by outliers, thus being a better representation of a typical salary.

Our point of comparison will be the annual median pay for the broader Computer and Information Technology (CIT) industry, which was $105,990 in 2024, reported by the U.S Bureau of Labor Statistics. This industry encompasses diverse areas such as computing theory, systems design, and practical technology applications, including Data Science itself, a specialized field within this domain.

## Setting Up The Hypothesis

We want to test whether the annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry.

- Null Hypothesis ($H_0$): The annual median salary for Data Scientists is equal to $105,990.
- Alternative Hypothesis ($H_A$): The annual median salary for Data Scientists is significantly higher than $105,990.
- The significance level is 0.05.

## Test Statistic

As a test statistic, the sample median summarizes the central tendency of the sample and is used to assess whether it significantly differs from the hypothesized population median (in this case, $105,990). It is particularly suitable for non-normally distributed data or when outliers are present, making it a robust alternative to the sample mean.

```
# Specify the null hypothesis
null_hypothesis <- data_scientist %>%
specify(response = salary_in_usd) %>%
hypothesize(null = "point", med = 105990)
```

## Null Distribution

To evaluate whether this observed difference is statistically significant, we generated a null distribution of sample median assuming the null hypothesis is true. As shown in the histogram below, the null distribution is approximately normal and centered around $105,990.

```
# Generate the null distribution
null_distribution <- null_hypothesis %>%
generate(reps = 10000, type = "bootstrap") %>%
calculate(stat = "median")
null_distribution
```

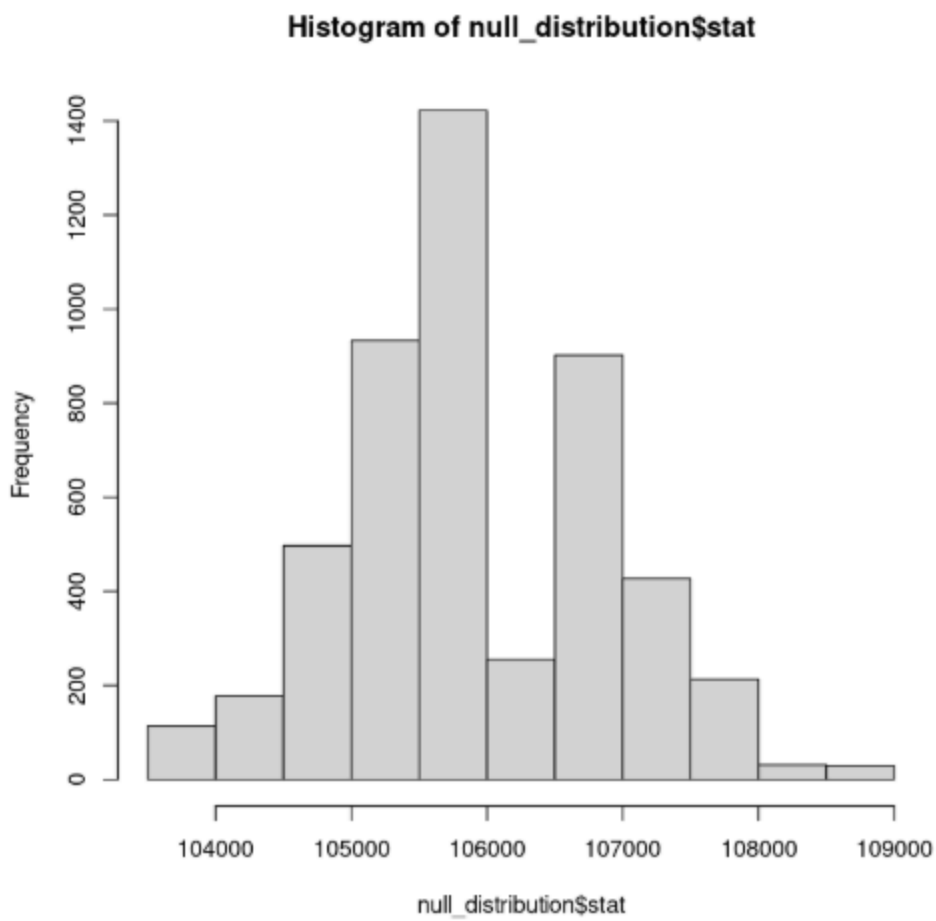

*Figure 1.1: Null distribution*

## Observed Test Statistic

We calculated the observed test statistic of the original sample to be $157,000.

```
# Calculate the observed statistic
 (observed_stat <- data_scientist %>%
 specify(response = salary_in_usd) %>%
 calculate(stat = "median"))
```

[17]

Table | Chart | Filter | Columns | Search

| Index | stat |
|---|---|
| 1 | 157000 |

Rows: 1 | Expand

Since the observed median of $157,000 lies far in the tail of this distribution, it would suggest that such a value is very unlikely under the null hypothesis, thus supporting the alternative hypothesis.

```
null_distribution%>% visualize()+ shade_p_value(obs_stat=observed_stat,
direction="greater")
```

[9]

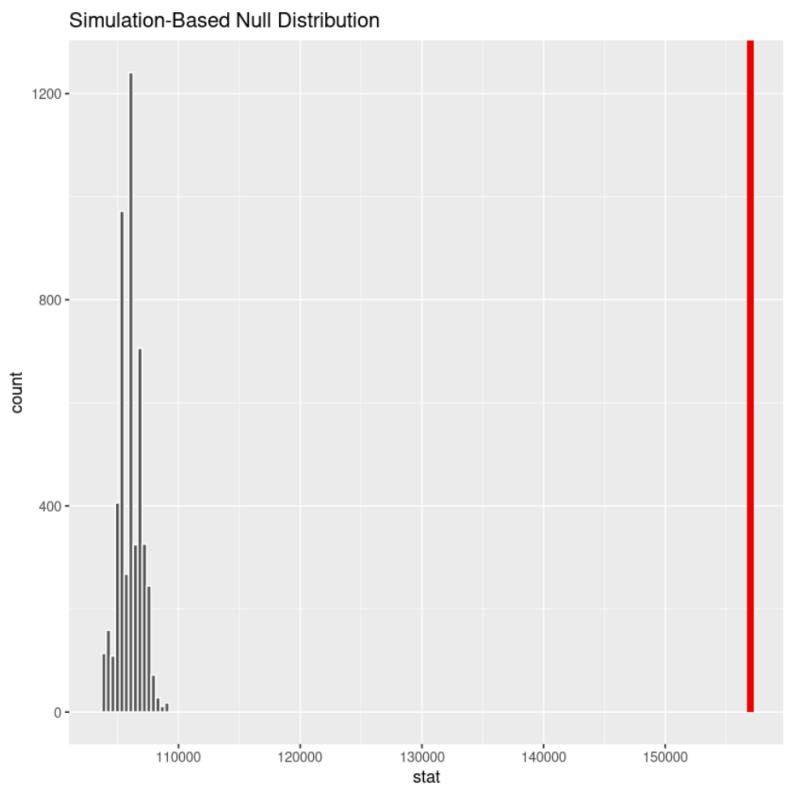


*Figure 1.2: Null distribution vs observed test statistic*

## Statistical Conclusion

To support the above observation, we calculated the p-value–the probability of the event that the null test statistic is more extreme than the observed value. The p-value is 0, smaller than our significance level. Thus, we have enough statistical evidence to reject the null hypothesis and support the alternative hypothesis that annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry.

```r
# Calculate the p-value
p_value <- null_distribution %>%
get_p_value(obs_stat = observed_stat, direction = "greater")
# Display the p-value
p_value
```

```
Warning message:
“Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
ℹ See `get_p_value()` (`?infer::get_p_value()`) for more information.”
```

Table | Chart | Filter | Columns | Search

| Index | p_value |
|---|---|
| 1 | 0 |

Rows: 1 | Expand

```r
# Set the significance level
alpha <- 0.05

# Assuming p_value is a numeric value, not a list or data frame
p_value <- 0.03  # Example p-value for demonstration

# Test conclusion
if (p_value < alpha) {
  conclusion <- "Reject the null hypothesis: The annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry. "
} else {
  conclusion <- "Fail to reject the null hypothesis: There is insufficient evidence to conclude that the annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry"
}

# Display the conclusion
conclusion
```

```
'Reject the null hypothesis: The annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry. '
```

## Bootstrap Confidence Interval

```r
boot_distn_one_median <- data_scientist %>%
  specify(response = salary_in_usd) %>%
  generate(reps = 10000, type = "bootstrap") %>%
  calculate(stat = "median")

ci <- boot_distn_one_median %>%
  get_ci()

ci

boot_distn_one_median %>%
  visualize(endpoints = ci, direction = "between") +
  shade_confidence_interval(endpoints = ci)
```

Table | Chart | Filter | Columns | Search

| Index | lower_ci | upper_ci |
|---|---|---|
| 1 | 155250 | 158755 |

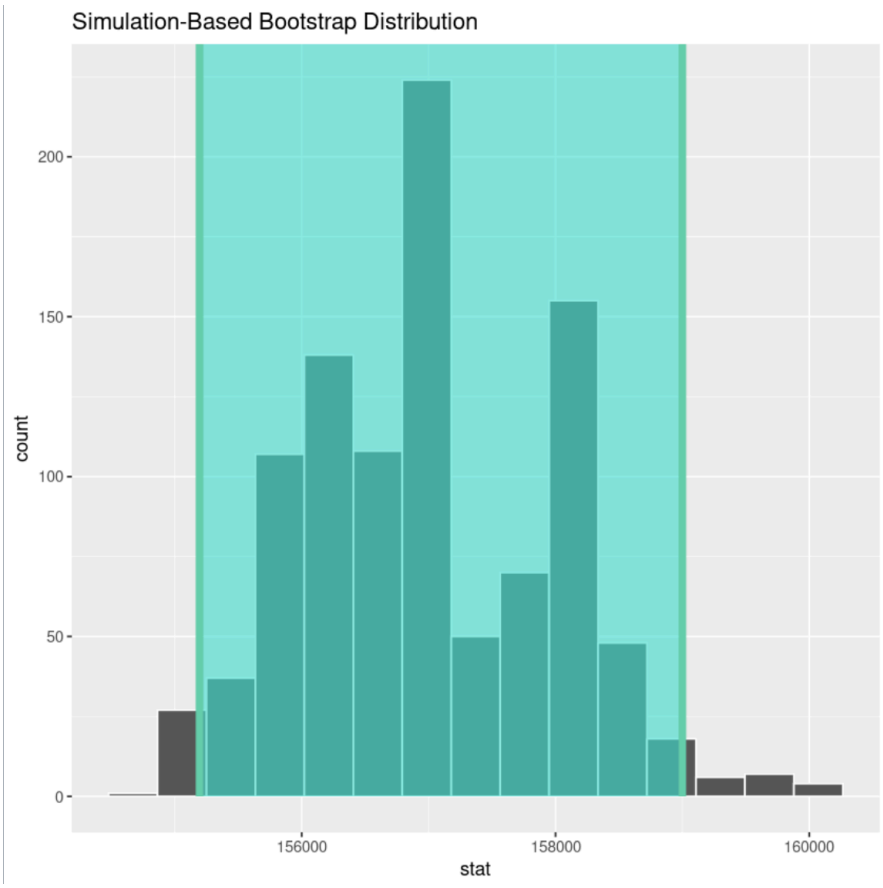


*Figure 1.3: 95% confidence interval*

The 95% confidence interval for annual median salary ranges from around slightly over $155,000 to $159,000, which is far higher than the hypothesized median $105,990. Therefore, we have enough statistical evidence to support the alternative hypothesis that the annual median salary for Data Scientists is significantly higher than the annual median salary for the CIT industry.

## Inference for a categorical variable

The decision to pursue a specific job role is often influenced by the desire for job stability. According to Matthew Di Carlo's analysis of the General Social Survey in the U.S between 1977 and 2014, many factors in the labor market affect the perceptions of job security. However, in our guidebook, we would only concentrate on the significance of full-time

employment market share as an indicator of job stability. This choice is grounded by a finding by Biscotini in 2024 about several benefits associated with full-time employment, including health insurance offered by the government, employment benefit packages, and higher chances of pay raises and promotion, overall allowing this employment type to be seen as stable and high-yield for workers.

# Literature Review

We found that in this field, full-time positions remain the dominant mode of employment. The U.S. Bureau of Labor Statistics (BLS) notes that “most data scientists work full time.” This aligns with survey data from the global data science community. For example, a Kaggle industry-wide survey (2017) found about 65% of data science and ML practitioners were employed full-time, compared to much smaller shares in non-traditional work modes (Part-time, Contract, Freelance).

## Setting Up The Hypothesis

Before setting the hypothesis, we performed an additional data cleaning step to standardize and group the values in the employment_type variable. Each entry in the employment_type column was examined as follows:

- If the value was “FT” (indicating Full-time employment), it was retained; all other values (e.g., “PT” for Part-time, “CT” for Contract, “FL” for Freelance, etc.) were recorded as "Others."

As a result, the employment_type variable is reduced to two categories: “FT” and “Others.” A frequency table was then generated to count the number of observations in each group, allowing for a clearer comparison of the distribution of full-time versus non-full-time employment types.

```
data$employment_type = ifelse( data$employment_type %in%
c("FT"),data$employment_type,"Others" )
table(data$employment_type)
```

```
    FT Others
 88111    473
```

Then, we set null hypothesis and alternative hypothesis:

- Null Hypothesis: The proportion of full-time employment is not higher than 65% ($H_0$: p≤0.65)
- Alternative hypothesis: The proportion of full-time employment is higher than 65% ( $H_A$:p>0.65)
- The significance level is 0.05.

## Test Statistic

The test statistic is the sample proportion.

The null distribution is the distribution of the test statistic when the null hypothesis is true.

```
# Specify hypotheses
null_hypothesis <- data %>%
  specify(response = employment_type, success = "FT") %>%
  hypothesize(null = "point", p = 0.65)
```

## Null Distribution

```
# Generate null distribution
null_distribution <- null_hypothesis %>%
  generate(reps = 500, type = "draw") %>%
  calculate(stat = "prop")
```

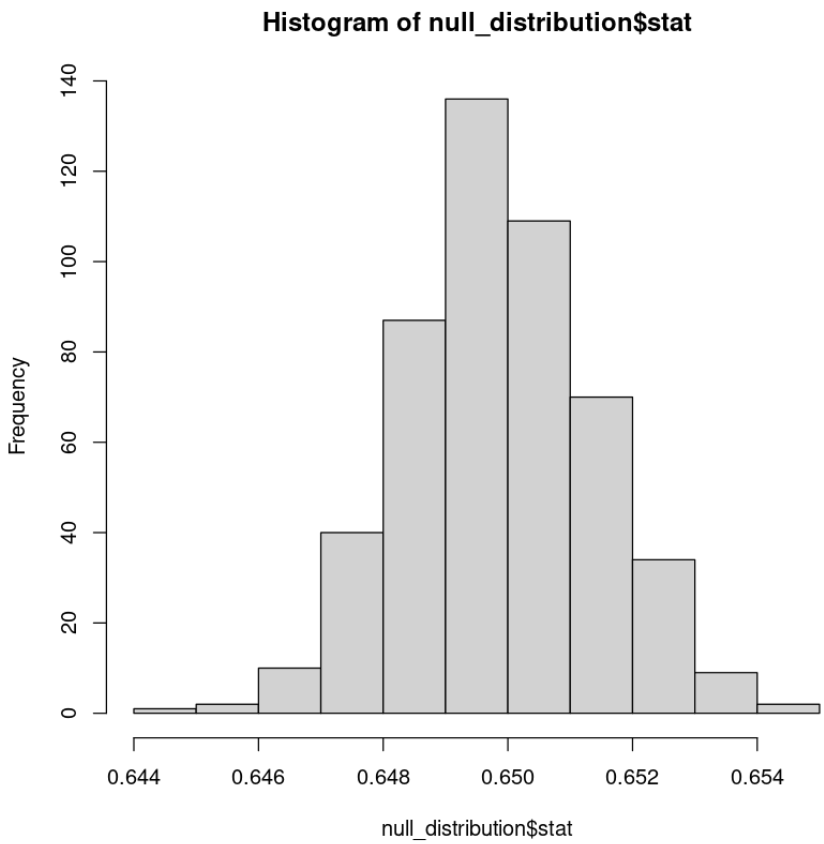


*Figure 2.1: Null distribution*

The null distribution of the proportion of full-time (FT) employees is generated by simulating samples under the assumption that the true proportion of FT workers is 0.65 (the null hypothesis). This distribution is approximately normal distribution, centered around 0.65, which reflects the value assumed under the null hypothesis.

## Observed Test Statistic

We calculated the observed test statistic of the original sample to be 0.9947.

```
observed_stat <- data %>%
  specify(response = employment_type, success = "FT") %>%
  calculate(stat = "prop")
observed_stat
```

Table | Chart | Filter | Columns | Search

| index | stat |
|---|---|
| 1 | 0.9947 |

*Figure 2.2: Observed data*

The observed proportion of 0.9947 lies far in the tail of the null distribution, indicating a high probability of rejecting the null hypothesis.

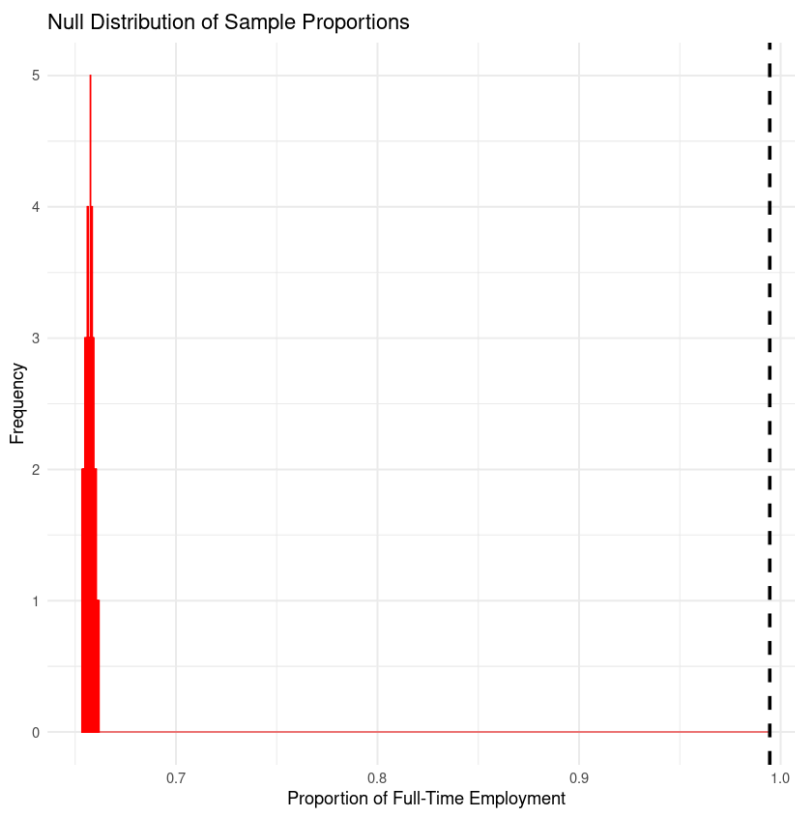


*Figure 2.3: Observed data compared to Null distribution*

## Statistical Conclusion

```r
# Calculate the p-value
p_value <- null_distribution %>%
  get_p_value(obs_stat = observed_stat$stat, direction = "greater")
```

| p_value |
| --- |
| 0 |

*Figure 2.4: P-value calculation*

```r
# Test conclusion
if (p_value$p_value < alpha) {
  conclusion <- "Reject the null hypothesis: There is strong evidence that The proportion
of full-time employment is higher than 65%."
} else {
  Conclusion <- "Fail to reject the null hypothesis: There is insufficient evidence to
conclude that The proportion of full-time employment is higher than 65%."
}

# Display the conclusion
conclusion
```

```
Warning message:
“Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
i See `get_p_value()` (`?infer::get_p_value()`) for more information.”
```

```
'Reject the null hypothesis: There is strong evidence that The proportion of full-time employment is
higher than 65%.'
```

To support the above observation, we calculated the p-value–the probability of event that the null test statistic is more extreme than the observed value. The p-value is 0, smaller than our significance level. Thus, we have enough statistical evidence to reject the null hypothesis and support the alternative hypothesis that the proportion of full-time employment is higher than 65%.

## Bootstrap Confidence Interval

We have further test with bootstrap confidence interval of values of test statistics:

```
boot_distn <- data %>%
  specify(response = employment_type, success = "FT") %>%
  generate(reps = 100, type = "bootstrap") %>%
  calculate(stat = "prop")

ci_boot <- boot_distn %>%
  get_ci()

ci_boot
```

Table Chart Filter Columns Search

| index | lower_ci | upper_ci |
|---|---|---|
| 1 | 0.9942 | 0.9951 |

```
boot_distn  %>%
  visualize(endpoints = ci_boot, direction = "between")
+shade_confidence_interval(endpoints=ci_boot)
```

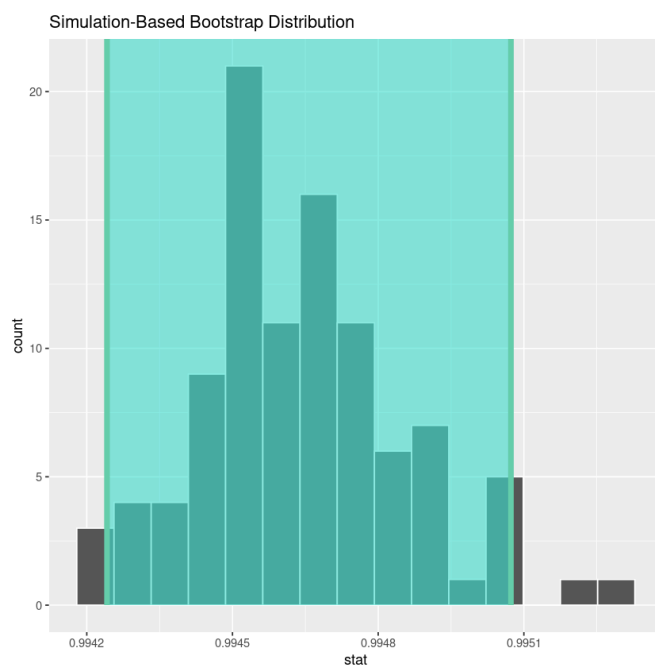


*Figure 2.5: 95% confidence interval*

The 95% confidence interval for the proportion of full-time employment ranges from 99,42% to 99.51%, which is higher than 65%. Therefore, we have enough statistical evidence to support the alternative hypothesis that the proportion of full-time employment is higher than 65%.

# Hypothesis test for dependence between two categorical variables: experience level and company size

Choosing an appropriate working environment is necessary for making significant strides in career development in the long-term. Conventionally, people would aim for large-scale companies in the hopes of better benefit packages and growth opportunities. However, given the cut-throat job market, it is important to be down-to-earth and think about whether our corresponding experiences are most sought after by which type of company.

## Literature Review

Taken together, the literature suggests a positive correlation between company size and the experience level of hires—with larger companies more likely to employ highly experienced professionals, and smaller firms relying more heavily on junior staff.

A growing body of research highlights that small and young tech firms disproportionately recruit junior talent. In particular, Burton et al. (2021) found that very small startups (approximately 12 employees) offered around 4% lower salaries to new hires compared to medium-sized firms (50+ employees), even for candidates with similar qualifications. Their study also revealed that startup employees tend to have less experience on average, and those who joined small firms early earned about 10% less over a ten-year period than their counterparts at larger companies.

These findings are consistent with broader labor market trends. According to Jahanshahi (2022) and Kepes (2024), small and emerging tech firms were responsible for the majority of tech job growth in the post-pandemic era, with non–Fortune 500 companies accounting for most new hires. These firms tend to favor entry-level and mid-level candidates, while often delaying or avoiding the hiring of very senior specialists.

One key explanation for this pattern is resource limitations. Small and medium-sized enterprises (SMEs) frequently face challenges in recruiting highly skilled professionals due

to lower wages, fewer benefits, and weaker brand recognition compared to larger firms (Benassi et al., 2021). For these companies, hiring junior employees is a cost-effective strategy that enables them to acquire affordable talent and develop it internally over time.

```
#Check the experience level segments in large companies
data_L <- data %>%
  filter(company_size == "L")

prop_table <- prop.table(table(data_L$experience_level))

prop_table

        EN         EX         MI         SE
0.13084112 0.04049844 0.42367601 0.40498442
```

```
#Check the experience level segments in medium companies
data_M <- data %>%
  filter(company_size == "M")

prop_table <- prop.table(table(data_M$experience_level))

prop_table

        EN         EX         MI         SE
0.04447745 0.01578989 0.30329086 0.63644180
```

```
#Check the experience level segments in small companies
data_S <- data %>%
  filter(company_size == "S")

prop_table <- prop.table(table(data_S$experience_level))

prop_table

        EN         EX         MI         SE
0.30952381 0.04761905 0.42857143 0.21428571
```

*Figure 3.1: Experience level in large, medium, small company*

Our dataset bears this trend out for the role of Data Scientist. Small companies have a much higher proportion of entry-level roles (roughly 30%) than large and medium-sized companies (about 13% and 4%, respectively), whereas large companies have a higher share of senior roles (roughly 40% and 63% at large and medium-sized firms vs ≈21% at small firms). In other words, there seems to be a correlation between experience level and company size: small companies tend to staff more juniors and fewer seniors than large ones.

## Setting Up The Hypothesis:

- Null Hypothesis *($H_0$):* Experience level and company size are independent.
- Alternative hypothesis ($H_A$): Experience level and company size are NOT independent.

- The significance level is 0.05.

## Test Statistic

We choose the test statistic Chi-square ($\chi^2$), defined as:

$$\chi^2 = \sum_{i=1}^{3}\sum_{j=1}^{4}\frac{(E_{ij}-O_{ij})^2}{E_{ij}},$$

where

- $E_{ij}$ is the *expected* count in the cell at row $i$, column $j$ under the null hypothesis that **company size** and **experience level** are independent;
- $O_{ij}$ is the *observed* count in the cell at row $i$, column $j$.

## Null distribution

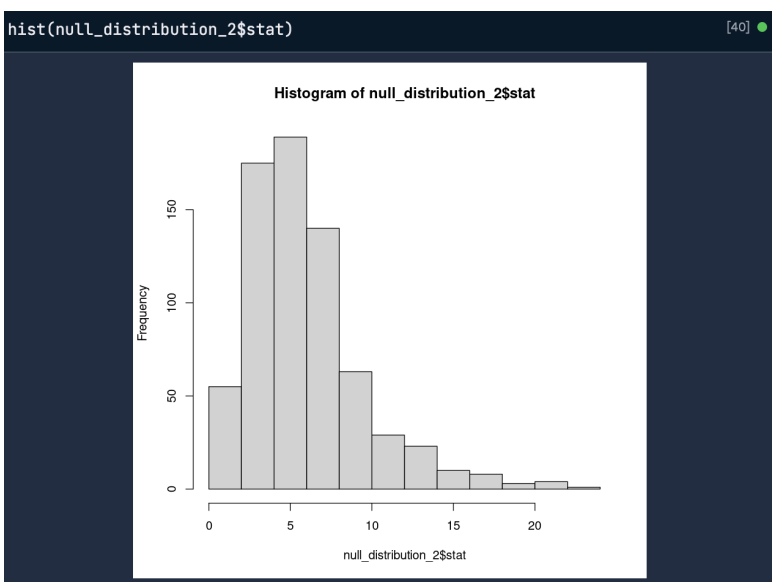


*Figure 3.2: Null distribution*

The null distribution of the Chi-square statistic $\chi^2$ was generated by repeatedly permuting the labels of *company size* and *experience level* to mimic the scenario in which the two variables are truly independent (the null hypothesis).

As the histogram shows, the simulated $\chi^2$-values are heavily right-skewed: most realisations fall between ≈ 2 and ≈ 10, with a long, diminishing tail that stretches to about 25. This shape is exactly what we expect from a $\chi^2$-distribution.

## Observed Test Statistic

```
# Calculate the observed statistic (Chi_square)
observed_stat_2 = sum((O_t-E_t)^2/E_t)

observed_stat_2
```

```
179.363318328377
```

*Figure 3.3: Observed data*

```
# Plot the null distribution

null_distribution_2 %>%
  visualize() +
  shade_p_value(obs_stat = observed_stat_2, direction = "right")
```

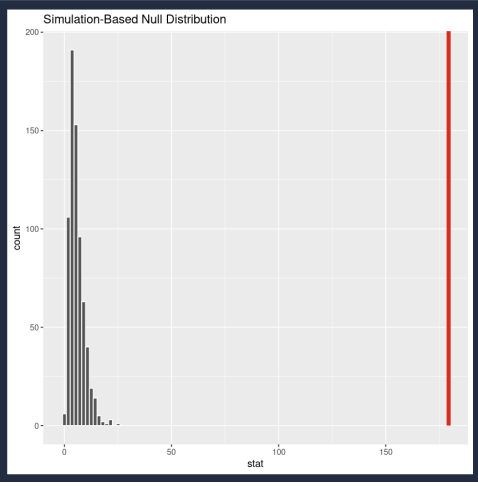


*Figure 3.4: Observed data compared to Null distribution*

When we add the observed Chi-square statistic, $\chi^2_{obs}$=179.36, on the permutation-based null distribution, it sits far out in the extreme right tail—well beyond the largest simulated value (~25). This enormous separation indicates that obtaining a statistic as large as 179.36 is virtually impossible if company size and experience level are truly independent.

```
# Calculate the p-value
(p_value <- null_distribution_2 %>%
  get_p_value(obs_stat = observed_stat, direction =  "greater"))

Warning message:
“Please be cautious in reporting a p-value of 0. This result is an approximation
based on the number of `reps` chosen in the `generate()` step.
ℹ See `get_p_value()` (`?infer::get_p_value()`) for more information.”
```

Table | Chart | Filter | Columns | Search

| index | p_value |
|---|---|
| 1 | 0 |

Rows: 1 | Expand

*Figure 3.5: P-value calculation*

Formally, the p-value—the proportion of resampled statistics at least as large as $\chi^2_{obs}$—is effectively zero. Thus, we have overwhelming evidence to reject the null hypothesis of independence, or in other words, the data provide strong support for the conclusion that company size and employee experience level are statistically associated.

## Bootstrap Confidence Interval

To validate the permutation test, we constructed a 95 % bootstrap confidence interval (CI) for the Chi-square statistic by repeatedly resampling the observed contingency table with replacement.

The resulting interval lies entirely above 11.929—the 95th-percentile cutoff of the null distribution—so the value 11.929 is *not* included in the CI. Because every plausible Chi-square value generated by the bootstrap procedure exceeds the threshold that would be expected under independence, we again conclude to reject the null hypothesis of

independence, and support the alternative hypothesis that company size and employee experience level are statistically associated.

```
# Plot the confidence interval
boot_distn %>%
  visualize(endpoints = ci_boot, direction = "between") +
  shade_confidence_interval(endpoints = ci_boot) + geom_vline(xintercept =
test_null_max, color = "red")
```

```
Warning message:
“The arguments `c("direction", "endpoints")` are deprecated in `visualize()` and
will be ignored. They should now be passed to one of `shade_p_value()` or
`shade_confidence_interval()`.”
```

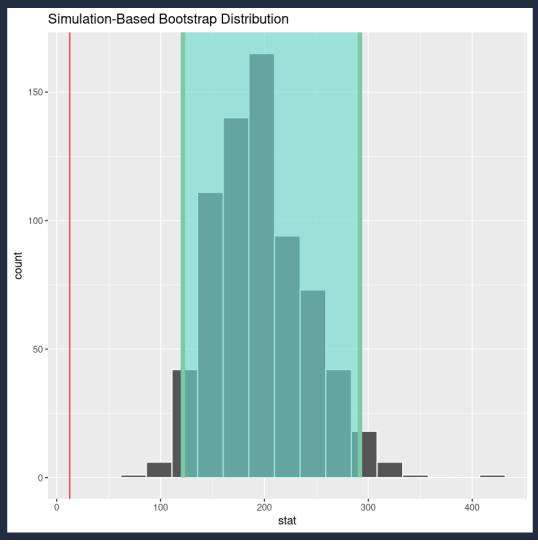


*Figure 3.6: 95% confidence interval*

# Hypothesis test for dependence between a numerical variable and a categorical variable: salary and remote ratio

In this section, we dive deeper into the effects of working status on salary levels, aiming to resolve concerns about whether the flexibility and comfort from remote working are associated with any negative trade-off in salary.

## Literature Review

There is an interesting finding by SHRM (a Human Resources agency in the US) that firms have to pay higher wages for the employees who work onsite to compensate for the high cost of living in the city centers–where most headquarters and officers are located. For example, software provider GitLab–an all-remote workforce, Facebook or Twitter cut the pay of employees opting to work away from their headquarters offices.

## Setting Up The Hypothesis

We want to test whether there is an actual difference in salary levels for these two different working modes.

Before stating the hypothesis, we start with cleaning data for this part. In the column "remote_ratio":

```
# Count each value in the 'remote-ratio' column
remote_ratio_counts <- data %>%
  count(`remote_ratio`)

remote_ratio_counts
```

[6]

Table Chart Filter Columns Search

| index | remote_ratio | n |
|---|---|---|
| 1 | 0 | 8991 |
| 2 | 50 | 12 |
| 3 | 100 | 2942 |

*Figure 4.1: Remote ratio counts*

- The number of entries for remote_ratio at 50 was disproportionately small compared to that of 0 and 100, only 12 out of the total number of 11,945 data points, indicating the insignificance of this type of working to the examined population. Thus, we group it together with the remote ratio 100 to represent the people working remotely.

As a result, the remote_ratio is now reduced to on-site work vs remote work.

- Null Hypothesis ($H_0$): There is no significant difference in the median salary levels for remote workers and on-site workers.

- Let $m_R$ be the median of salary levels for remote workers.
- Let be $m_O$ the median of salary levels for on-site workers.
- $H_0$: $m_R = m_O$.
- Alternative Hypothesis ($H_A$): There is a significant difference in the median of salary levels for remote workers and on-site workers.
- $H_A$: $m_R \neq m_O$.
- The significance level is 0.05.

## Test Statistic

The test statistic is the difference between the two sample medians: $m_R - m_O$.

The null distribution is the distribution of the test statistic when the null hypothesis is true.

The two-sided difference between two sample medians (median_remote - median_on_site), means that when the sign of difference is negative (median_remote - median_on_site = -value), the salary levels for remote employees are lower than those of on-site employees, vice versa.

## Null Distribution

To evaluate whether this observed difference is statistically significant, we generated a null distribution of difference between two sample medians assuming the null hypothesis is true. As shown in the histogram below, the null distribution is approximately normal and centered around 0.

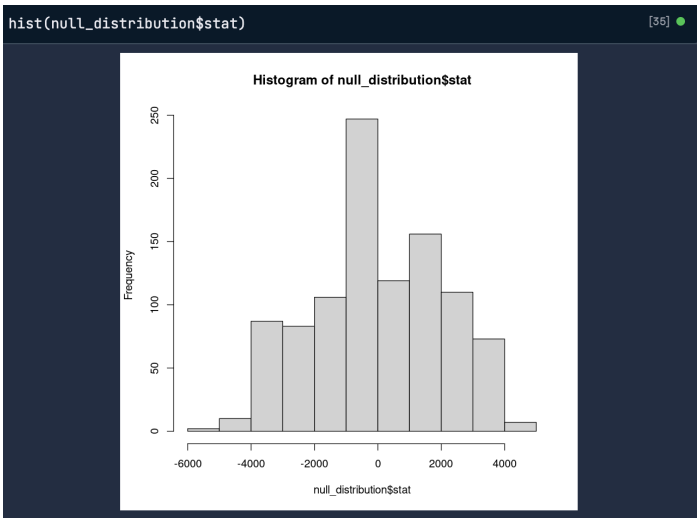


*Figure 4.2: Null distribution*

## Observed Test Statistic

We calculated the observed test statistic of the original sample to be **-5200**.

```
(observed_stat <- data %>%
    specify(salary_in_usd ~ remote_ratio)  %>%
  calculate(stat = "diff in medians", order = c("remote", "on_site")))
```

[19]

Table | Chart | Filter | Columns | Search

| index | stat |
|---|---|
| 1 | -5200 |

*Figure 4.3: Observed data calculation*

This means there is an actual difference in salary levels between remote employees and on-site employees.

```
null_distribution%>% visualize()+ shade_p_value(obs_stat=observed_stat,
direction= "both")
```

[37]

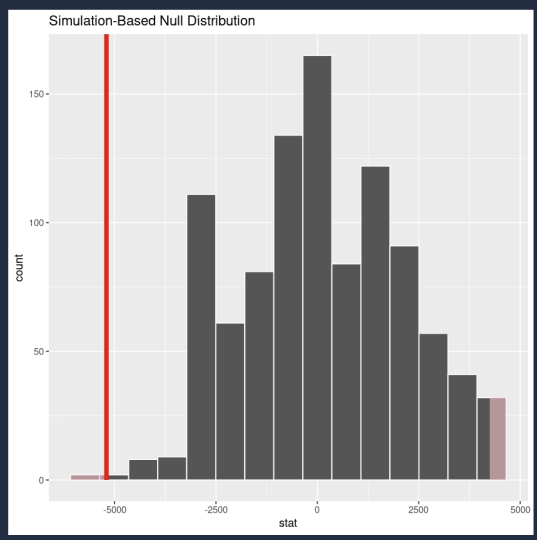


*Figure 4.4: Observed data compared with null hypothesis*

When adding the observed data "**-5200**" in the distribution, we witness a gap between the null hypothesis and the observed data, indicating a high probability of rejecting the null hypothesis. In order words, There is a significant difference in the median salary for remote ratio.

## Statistical Conclusion

```
(p_value <- null_distribution %>%
  get_p_value(obs_stat = observed_stat, direction =  "both"))
```

[39]

Table | Chart | Filter | Columns | Search

| index | p_value |
|---|---|
| 1 | 0.004 |

*Figure 4.5: P-value calculation*

In order to conclude, we calculate the p-value which is "**0.004**" (smaller than 0.05) meaning that we have enough statistical evidence to reject the null hypothesis and support the

alternative hypothesis: There is a significant difference in the median salary for remote status.

## Bootstrap Confidence Interval

Our conclusion is also enhanced by the 95% confidence interval:

```
ci <-boot_distn  %>%
  get_ci()
ci
```

[44]

Table | Chart | Filter | Columns | Search

| index | lower_ci | upper_ci |
|---|---|---|
| 1 | -10000 | -1500 |

*Figure 4.6: 95% confidence interval calculation*

```
boot_distn %>%
  visualize(endpoints = ci, direction = "between")
+shade_confidence_interval(endpoints=ci)
```

```
Warning message:
“The arguments `c("direction", "endpoints")` are deprecated in `visualize()` and
will be ignored. They should now be passed to one of `shade_p_value()` or
`shade_confidence_interval()`.”
```

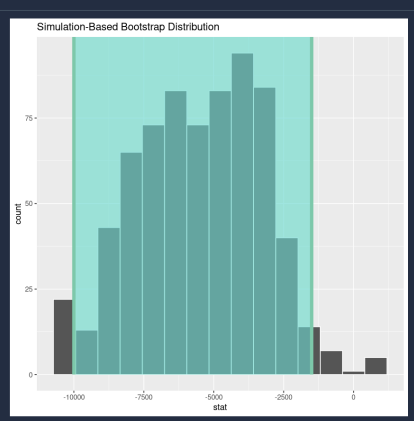


*Figure 4.7: 95% confidence interval*

The 95% confidence interval for the difference in the median of salary levels for remote workers and on-site workers  ranges from -10000 to approximately -1500, which is far below 0. Therefore, we do have enough statistical evidence to make a conclusion of no

difference in salary levels for two groups, then it supports the alternative hypothesis that there is statistically significant difference between remote salary and onsite salary.

To identify which group yields higher annual salary levels, we consider the sign of the difference. Since the observed data is -5200, we can conclude that remote workers earn lower salaries compared to on-site workers.

```
library(ggplot2)                                                    [42]

# Combine the two bootstrap distributions into one data frame for plotting
boot_distn_combined <- rbind(
  data.frame(group = 'on_site', stat = boot_distn_on_site$stat),
  data.frame(group = 'remote', stat = boot_distn_remote$stat)
)

# Plot the histogram
p <- ggplot(boot_distn_combined, aes(x = stat, fill = group)) +
  geom_histogram(position = 'identity', alpha = 0.5, bins = 30) +
  labs(title = 'Bootstrap Distributions of Salary (USD)',
       x = 'Bootstrap Mean Salary (USD)',
       y = 'Frequency') +
  theme_minimal()

print(p)
```

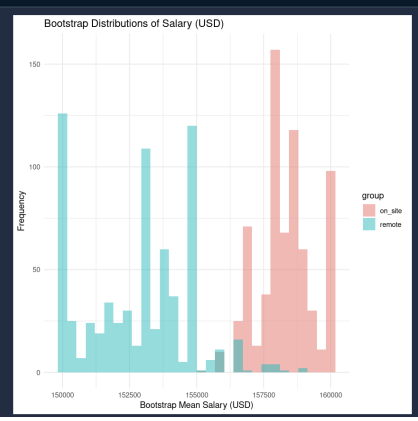


*Figure 4.8: Bootstrapping median salary in two groups*

## Conclusion & Discussion

### Conclusion:

This analysis highlights key factors shaping the landscape of Data Science careers in the U.S, offering practical guidance for Math graduate students considering this path.

First and foremost, compensation is a strong incentive—Data Scientist roles command a median annual salary significantly higher than the median pay for roles in the broader tech sector. Employment type also plays a critical role. Over 65% of positions are full-time,

indicating a substantial degree of job stability, which is encouraging for graduates seeking secure, long-term career opportunities.

Having set their hearts on pursuing a career in this field, graduates should then choose an appropriate working environment according to their experience level. This advice is well-informed by our test, which proves that there is a significant relationship between company size and experience level: smaller firms are more inclined to hire entry-level talent, while larger firms tend to employ more senior professionals. This suggests a strategic pathway—start at smaller companies to build experience, then transition to larger firms for leadership or specialized roles.

Lastly, based on the conclusion that there is a negative difference between median salary for remote workers and on-site workers respectively, we have gained an insight that working away from offices leads to deduction in salaries. However, the decision to choose either of them would depend on personal preferences and demands, as a result indicates that even if there is a deduction in salaries for remote workers, they are still content with that for several benefits, including more balanced life time flexibility.

**Discussions:**

To build a more comprehensive understanding of the field, future research could explore:

- Geographic and economic context – How do regional job markets and cost-of-living variations affect salary competitiveness and job accessibility in different parts of the U.S?
- Career trajectory and mobility – What factors (e.g., certifications, project portfolios, company changes) influence the pace and direction of career advancement in Data Science?
- Remote work implications – How does remote flexibility relate to job satisfaction, performance, and salary trends in data-driven roles?
- Diversity and equity in hiring – How do gender, race, and educational background impact access to Data Science opportunities and long-term outcomes?
- Dataset scope and accuracy – Future studies should increase the sample size, diversify data sources, and validate the collection timeline to ensure findings remain representative and current.

By addressing these directions, Math graduates can better navigate the evolving job market and make data-informed decisions in the high-potential, fast-growing world of Data Science.

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## Data Inference

- Part 1: Inference for a numerical variable
- Part 2: Inference for a categorical variable
- Part 3: Hypothesis test for dependence between two categorical variables: experience level and company size
- Part 4: Hypothesis test for dependence between a numerical variable and a categorical variable: salary and remote ratio